Lei Ding

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EDUCATION

University of California, Santa Cruz Aug. 2022 ~ Now
Computer Science Ph.D.: Agentic Post-Training; Harness–Model Co-Evolution Santa Cruz, California, U.S.
Sichuan University Aug. 2004 ~ July. 2007
Master: Applied Computer Technology - Computer Network Sichuan, China
Sichuan University Aug. 2000 ~ July. 2004
Bachelor: Software Engineering Sichuan, China

PUBLICATIONS

- Lei Ding, Bin He, Brian Quanz, Andrew P Martin, Jyoti Prakash Maheswari, Chenguang Wang, Yang Liu, TITRATE: Adaptive Coordination of Skill Evolution and Policy Learning in Agentic Self-Distillation, ICLR 2027 (In review)
- Xiwen Wei, Bin He, Brian Quanz, Jyoti Prakash Maheswari, Lei Ding, MO ZHANG, Dhanashri Prashant Tidke, Andrew P Martin, Turing2 - Scaling Turing-Style Evaluation to Optimize Human Simulation in Dialogue, ICLR 2027 (In review)
- Lei Ding, Chris Liu Haoyu, Chenguang Wang, Yang Liu, Task Synthesis and Unified Web Agent Training Framework on Open Domain. (Ongoing)
- Lei Ding, Zonglin Di, Chenguang Wang, Yang Liu, Dynamic MCTS Tree Using "Plan-Act-Critic" Multi-Agent Framework. (Ongoing)
- Yiyou Sun, Xinyang Han, Weichen Zhang, Lei Ding, etc., Agents' Last Exam, NeurIPS 2026
- Lei Ding, Bin He, Chenguang Wang, Yang Liu, ProActor: Timing-Aware Reinforcement Learning for Proactive Task Scheduling Agents. ACL 2026 & ICLR 2026 Agents in the Wild Workshop.
- Vanshika Vats, Lei Ding, James Davis, et al., A Survey on Human-AI Collaboration with Large Foundation Models. ACM Transactions on Intelligent Systems and Technology, Volume 18, Issue 1, 2026.
- Wendy Yaqiao Liu, Rui Jerry Huang, Anastasia Miin, Lei Ding, Adaptive Coopetition: Leveraging Coarse Verifier Signals for Resilient Multi-Agent LLM Reasoning. NeurIPS 2025, IJCNLP-AACL 2025.
- Li Liu, Diji Yang, Sijia Zhong, Kalyana Suma Sree Tholeti, Lei Ding, Yi Zhang, Leilani H. Gilpin, Right this way: Can VLMs Guide Us to See More to Answer Questions? NeurIPS 2024.
- Timothy Wei, Hsien Xin Peng, Elaine Xu, Bryan Zhao, Lei Ding, Diji Yang, Dual-Model Distillation for Efficient Action Classification with Hybrid Edge-Cloud Solution. NeurIPS 2024.
- Lei Ding, Yue Fan, et al., Read Anywhere Pointed: Layout-aware GUI Screen Reading with Tree-of-Lens Grounding. EMNLP 2024.
- Lei Ding, Yi Zhang, Jeshwanth Bheemanpally, Enhancing Mobile" How-to" Queries with Automated Search Results Verification and Reranking. The Second Workshop on Generative Information Retrieval, SIGIR 2024.
- Wu Weiping, Ding Lei. Digitalization and upgrading of drug design and development by artificial intelligence. Changsha, international peptide drugs and innovation summit 2022, oral presentation.
- Abena AChiaa Atwereboannah, Wu Weiping, Ding Lei, Sophyanbi B. Yussif, Edwin Tenagyei. Protein-ligand binding affinity prediction using Deep Learning, 2021 18th International Computer Conference on Wavelet Active Media Technology and Information Processing (ICCWAMTIP 2021), 56.

PATENTS & SOFTWARE COPYRIGHT

- Co-Evolution of Model Parameters and an External Skill Harness, US Provisional Patent Application (filed) | Zillow Ref. P260058USP1, 2026
- Timing-Aware Reinforcement Learning for Proactive Task-Scheduling Agents, US Patent Application (filed) | Zillow Group, 2025
- Rating of city road segments for taxi hailing based on HANA technology, US Application NO. 13/934,706 | Patent ID 81495268 | Patent Ref 120542US01, China Application NO. 20130269463.3 | Patent ID 82826027 | Patent Ref 120542CN01

- Automatic category assignment and potential topic discovery for products based on Latent Dirichlet Topic algorithm, SAP Patent Invention ID. 83839165
- Simulator of bundle clicking for validating Bandit strategies in A/B testing, US Application NO. 17/547,637 | Patent ID 83839171 | Patent Ref 210412US01
- Reinforcement Learning Model for product recommendation considering balance between product profit and customer interests, US Application NO. 17/556,238 | Patent ID 83848635 | Patent Ref 210416US01

RESEARCH

University of California, Santa Cruz 2022.08-Present

Computer Science Ph.D. Candidate

Research Focus: Agentic post-training of LLM agents (reinforcement learning and on-policy self-distillation), centered o n harness–model co-evolution: the agent harness (skills, tools, runtime) and the model weights are updated in one close d loop, so that experience refines the harness and the harness in turn guides policy learning.

- **Harness–Model Co-Evolution via Skill-Augmented Self-Distillation (TITRATE)**: Designed and developed a co-evolution framework in which rollout experience refines a versioned skill bank and the skills in turn guide policy learning through on-policy self-distillation, coordinated by an evidence-gated, budget-aware REUSE/REFRESH controller and a progress-aware distillation schedule to train stronger agents at a fraction of the analyzer cost—with future work extending co-evolution from skills to the full agent harness.
- **Task Synthesis and Unified Web Agent Training Framework on Open Domain**: Designed and developed an end-to-end web agent training framework that synthesizes verifiable tasks on live open-domain websites via explorer–verifier agents, trains small open models with GRPO and hindsight-skill self-distillation on a shared Playwright browser stack, and delivers strong out-of-domain transfer that closes much of the gap to frontier models.
- **End-to-End Proactive Agent Reinforcement Learning Framework**: Designed and developed a full-stack RL training system that defines fine-grained proactiveness metrics, integrates turn-level GRPO with RULER-enhanced rewards, and leverages the optimized ART-F asynchronous RL infrastructure to enable efficient, timing-aware learning for task-scheduling agents—even on quantized LoRA models.
- **Dynamic MCTS with a Plan-Act-Critic Multi-Agent Architecture**: Designed a dynamic MCTS framework that decomposes long-horizon GUI tasks into subtasks, continuously refines plans, and improves exploration efficiency through node reweighting, action relocation, and critic-driven evaluation to boost GUI agent performance.
- **Optimizing GUI Agent Behavior with Layout-Aware Perception:** Enhance GUI agents by improving how they perceive, select, ground, and execute actions using layout-aware visual understanding, hierarchical reasoning, and critic-informed feedback to produce more accurate, efficient, and reliable interaction trajectories.
- **Agent-Driven Web Search Optimization**: Investigate how execution-capable AI agents can verify step-by-step "How-to" solutions extracted from search results and transform execution feedback into stronger reranking signals. This includes plan extraction from webpages, building LLM-based acting agents to complete tasks, and designing reranking models grounded in real execution outcomes.

WORKING EXPERIENCE

Zillow Group 2026.06-2026.11

Applied Science Intern (Returning), AI Research

- **Harness–Model Co-Evolution (Co-Harness-RL):** Designed and developed TITRATE, a training recipe that co-evolves an external skill harness with model weights for long-horizon, sparse-reward LLM agents via evidence-gated skill refresh and adaptive on-policy self-distillation (ICLR 2027 submission).
- **Patent:** Co-inventor of a US provisional patent application, "Co-Evolution of Model Parameters and an External Skill Harness".
- **Human Simulation for Assistant RL:** Contributed to Turing2, scaling Turing-style evaluation to optimize human simulation in dialogue for assistant training and evaluation (ICLR 2027 submission).

Zillow Group 2025.06-2025.09

Applied Science Intern, AI Research

- **Proactive Agent RL (ProActor):** Designed and developed a timing-aware RL framework for proactive task-scheduling agents with turn-level GRPO, RULER-enhanced rewards, and the ART-F multi-GPU asynchronous RL infrastructure (4–8× training speedup), published at ACL 2026.
- **Patent:** Co-inventor of a US patent application on timing-aware reinforcement learning for proactive agents.

SAP Upscale, SAP Labs 2017.07-2022.08

Senior Data Scientist and Architect

- Conducted convergence analysis of A/B testing of product bundles (product combination) that customers show decayed interest in using Epsilon Greedy, Softmax and UCB1 Bandit Algorithms for customer behaviors' simulation.
- Analyzed product similarity and complement relationship via Latent Dirichlet Allocation Model given the products category tree and the text feature of products.
- Discover and generate product bundle via Collaborative Filtering, Apriori, FP-Growth based on short-term customer interests.
- Generated dynamic product bundles and recommended product items to customers using Deep Reinforcement Learning in order to hit the balance between gaining product profit for merchant and fulfillment of customer interests based on product features and short-term customers' behavior data.

SAP Engagement Center on Cloud Infrastructure, SAP Labs 2015.04-2017.06

Senior Software Engineer and Architect

- Conducted service exception discovery model based on Multivariate Gaussian distribution and analyzed efficiency of exception handling in the system.
- Analyzed system bottleneck and optimized services based on payload statistics and service dependency graph.

Big data application and algorithm optimization in SAP Nanjing Innovation Center, SAP Labs 2011.04-2015.03

Algorithm Lead and Architect

- Optimized CONOP[1] based on Simulated Annealing Algorithm to determine relative time scale of fossil records, and proposed a parallelization solution based on Monte Carlo sampling: a co-innovation project with Nanjing Institute of Geology and Paleontology[2]
- Designed the Nanjing Smart Traffic Platform, including Origin-Destination analysis, city congestion analysis, dynamic traffic zone extraction, short-term congestion prediction, fake vehicle plate number discovery.

Platform development and partner toolkit development for SAP Business ByDesign, SAP Labs 2007.05-2011.03

Algorithm and Application Developer

- Developed Business Object Description Language (BODL) and Advanced Business Scripting Language (ABSL) based on ANTLR, and integrated them with Eclipse and Visual Studio for SAP Partner Development Infrastructure(PDI).
- Developed Visual Studio plugins and visual editor of User Interface (UI) by an across-AppDomain communication framework that also guarantees process security via .Net AppDomain isolation.

HONOR & CERTIFICATES

- 2003 Microsoft Innovation Cup - SALT, The school team, 12th place of Mainland China, Touronline (virtual tour online)
- 2010 SAP Excellent Employee，2012 SAP High Potential Employee
- 2013 and 2014 Team Coach of Nanjing Innovation Center in SAP Innovation Competition, 1st place of China Lab
- Tianchi CIMKM AnalytiCup 2018, Cross-lingual Short-text Matching of Question Pairs, 26/1027
- Coursera Andrew Ng's Machine Learning Series(with programming assignment)，refer to certificate
- Coursera Andrew Ng's Deep Learning Series (with programming assignment), refer to certificate
- Coursera Daphne Koller's Probabilistic Graph Model 1，2，3(with programming assignment)，refer to certificate
- Coursera Geoffrey Hinton's Neural Networks for Machine Learning, refer to certificate
- Udacity Michael Littman and Charles Isbell's Reinforcement Learning
- UC Berkeley CS285 by Sergey Levine，refer to programming assignment

[1] CONOP, refer to https://www.paleosoc.org/assets/docs/extended-CONOP-COURSE-NOTES.pdf

[2] Algorithm design, refer to SAP China Tech Talk ZE038 delivered by Orlando